

Comprehensive Project Report (Capstone Dissertation)

Final Engineering Report on AI-Based Household Appliance Rental Platform

Academic and strategic research artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2). Demonstrating circular economy transition, machine learning churn mitigation, FunkSVD recommendation, and localized generative AI edge deployment.

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Principal Authors:	Advanced AI & Systems Research Group	Affiliation:	Department of Computer Science & Engineering
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Tech / M.Tech Senior Engineering Capstone Project

BONAFIDE CERTIFICATE: This is to certify that the project report entitled "*AI-Based Household Appliances Rental Platform (RentAI / SDC2)*" is a bona fide record of work carried out under university academic supervision and meets all engineering degree criteria.

Chapter 1: Introduction & Domain Motivation

With accelerating urbanization in metropolitan tech centers, the demand for flexible, asset-light living has outpaced traditional retail models. RentAI addresses this paradigm shift by delivering a unified, scalable software platform where home appliances are offered as a dynamic, service-managed subscription.

Chapter 2: System Architecture & NoSQL Design

The platform is architected around a pure NoSQL data layer utilizing **MongoDB 8.3**. By decoupling schema definitions from rigid relational table structures, RentAI indexes diverse appliance hardware attributes (dimensions, energy ratings, defrost mechanisms) across 1,433 units while sustaining sub-15ms query response times.

Chapter 3: Machine Learning & Predictive Engines

3.1 LightGBM Customer Churn Mitigation

The churn pipeline operates as a real-time risk diagnostic system. By training on 15 behavioral telemetry features with 5-fold cross-validation, the model achieves a test accuracy of **95.67%**, an ROC-AUC of **0.9893**, and a low log loss of **0.1189**.

3.2 Tri-Paradigm Recommender

The recommendation subsystem delivers customized inventory recommendations via Biased FunkSVD latent matrix factorization ($k=20$), supported by User-Based and Item-Based cosine similarity fallbacks.

Chapter 4: Local RAG Chatbot with Ollama

Rather than relying on costly third-party cloud APIs that compromise privacy, RentAI embeds a local Ollama supervisor on port 11434. The assistant executes keyword-driven Retrieval-Augmented Generation against live MongoDB inventory records, answering user inquiries within 1.8 seconds on standard consumer CPUs.

Chapter 5: Verification, Benchmarks & Conclusion

Automated test suites confirm 100% pass rates across all 9 API gateways. The integrated system demonstrates enterprise reliability, zero-cost edge AI operation, and exceptional machine learning predictive accuracy.